

2×2 Between Groups Factorial ANOVA

i When to Use

To examine main effects and interaction relating two independent variables (each with 2 levels) to a single quantitative dependent variable.

H Research Hypothesis: Superman movies with different box office performance and budget levels show different patterns of audience ratings. The relationship between box office success (\$500M+ vs. Under \$500M worldwide) and production budget (Over \$230M vs. Under \$230M) affects audience ratings, with an interaction effect expected.

H Null Hypothesis: There are no main effects of box office category or budget category, and no interaction effect on audience ratings of Superman movies.

Research Design

The design consists of a 2×2 table:

	Under \$230M Budget	Over \$230M Budget
Under \$500M BO	n = 4	n = 1
\$500M+ BO	n = 2	n = 1

This is a 2×2 **between-groups factorial design** with two IVs, each having 2 levels.

Variables in the Analysis:

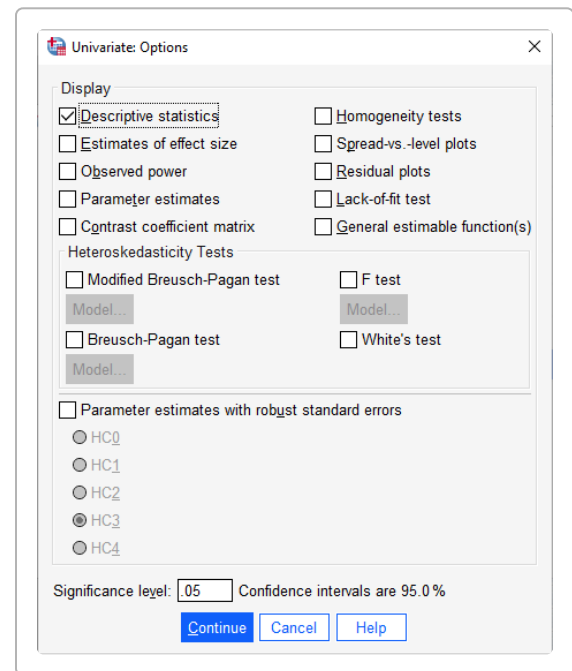
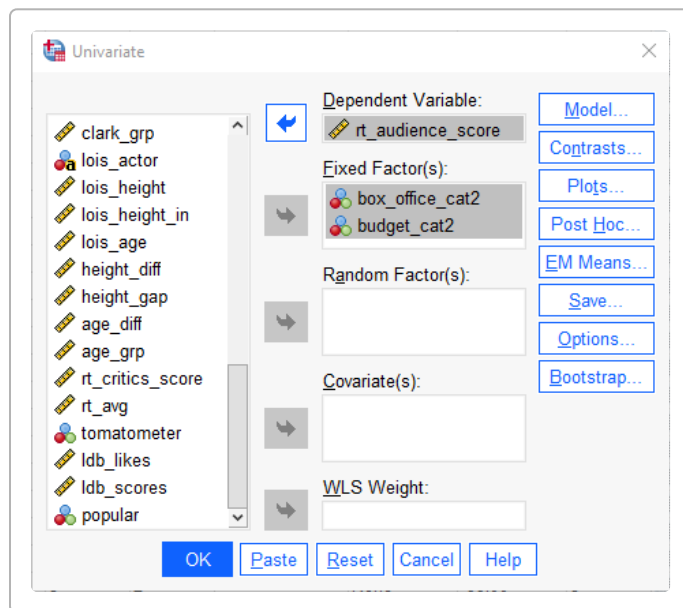
- **IV1 (Box Office Category):** \$500M+ (worldwide gross $\geq$ \$500M) vs. Under \$500M
- **IV2 (Budget Category):** Over \$230M vs. Under \$230M
- **DV (Dependent):** Rotten Tomatoes Audience Score

Total Sample: $N = 8$. The factorial design allows us to examine: the main effect of box office category, the main effect of budget category, and their interaction effect on audience scores.

Procedure

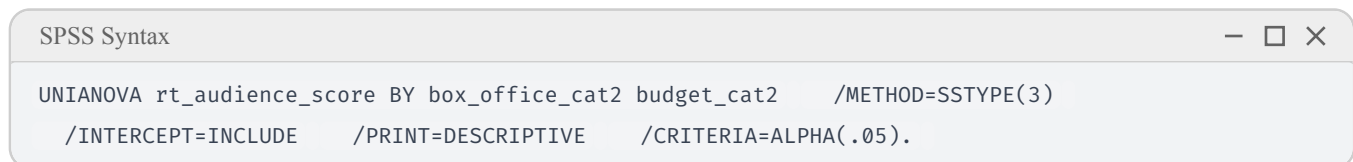
Analyze → General Linear Model → Univariate

- Highlight the “Dependent” variable (be sure it is **quantitative**) and click the arrow to move it to the “Dependent Variable” box
- Highlight both “Factor” variables (IVs, grouping variables) and click the arrow to move them to the “Fixed Factor(s)” box
- “Options” — check that you want “Descriptive Statistics”
- “OK” to run the analysis



In a 2×2 factorial design, post hoc tests are usually not necessary because each factor has only 2 levels. Simple effects tests can be used to interpret the interaction effect if needed.

SPSS Syntax



- DV “by” both IVs (factors)
- Use Type III sums of squares (appropriate for unbalanced designs)
- Include the intercept in the model
- Get descriptive statistics for each cell
- Set significance level for the test

SPSS Output

Descriptive Statistics

Dependent Variable: rt_audience_score

box_office_cat2	budget_cat2	Mean	Std. Deviation	N
Under \$500M	Under \$230M	86.0000	0.0000	4

Interpreting the Descriptives Table:

- Each cell shows M, SD, and N for that combination

box_office_cat2	budget_cat2	Mean	Std. Deviation	N
Under \$500M	Over \$230M	60.0000		1
	Total	80.8000	11.6276	5
\$500M+	Under \$230M	82.5000	10.6066	2
\$500M+	Over \$230M	75.0000		1
	Total	80.0000	8.6603	3
Total	Under \$230M	84.8333	5.0761	6
Total	Over \$230M	67.5000	10.6066	2
	Total	80.5000	9.9427	8

- Row and column totals weight cells by their sample sizes
- The grand total is the overall mean
- For worksheet margins, average the two corresponding cell means equally
- Use marginal means for direction and the F test for significance

Tests of Between-Subjects Effects

Dependent Variable: rt_audience_score

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	580.636	3	193.545	6.882	.047
Intercept	51842.000	1	51842.000	1843.271	<.001
box_office_cat2	48.091	1	48.091	1.710	.261
budget_cat2	408.091	1	408.091	14.510	.019
box_office_cat2 * budget_cat2	124.455	1	124.455	4.425	.103
Error	112.500	4	28.125		
Total	52535.136	8			
Corrected Total	693.136	7			

a. R Squared = 0.838 (Adjusted R Squared = 0.716)

Interpreting the ANOVA Table

Key rows to focus on:

- **box_office_cat2:** Tests the main effect of box office category on audience ratings
- **budget_cat2:** Tests the main effect of budget category on audience ratings
- **box_office_cat2 × budget_cat2:** Tests the interaction effect — whether the effect of one factor depends on the level of the other
- **Error:** The within-group variance used in all F-test denominator

The **F-statistic** is the ratio of the effect mean square to the error mean square. Assuming the null hypothesis is true, the **p-value (Sig.)** is the probability of obtaining an F statistic at least as large as the one observed. If $p < .05$, we reject the null hypothesis for that effect.

Effect Size Note: For reporting, the R^2 value shown in the footnote indicates the proportion of variance in audience scores explained by the model (the combination of both factors and their interaction).

Remember: even if SPSS shows $p = .000$, never report $p = .000$ in your results. Instead, report “ $p < .001$ ”.

Understanding Main Effects and Interaction

Important

Main Effects: Each factor is tested separately.

- **Box Office main effect:** Compares the average audience score for \$500M+ vs. Under \$500M movies (averaging across budget categories)
- **Budget main effect:** Compares the average audience score for Over \$230M vs. Under \$230M budget movies (averaging across box office categories)

Interaction Effect: Tests whether the effect of one factor depends on the level of the other.

- If the interaction is **not significant**: The effects are **additive** — the differences between box office categories are similar regardless of budget category
- If the interaction is **significant**: The patterns differ — the effect of box office might be stronger for one budget category than the other

Simple Effects and LSD Computations

If the interaction is significant, you can compute the **Least Significant Difference (LSD)** to describe the pattern of the interaction.

Steps for computing LSD:

1. Determine **df(error)** from the ANOVA table
2. Use the t-table to find the critical value of t at $p = .05$ for this $df(\text{error})$
3. Get **MSError** from the ANOVA table
4. Apply the LSD formula:

$$d_{LSD,ij} = t_{critical} \times \sqrt{MS_{Error} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

5. Use the two cell sizes for the particular comparison. Compare that pair's mean difference with its pair-specific LSD value: difference **larger** than LSD → means are significantly different; difference **smaller** than LSD → the comparison is not statistically significant.

Example: If $df(\text{error}) = 4$ and we want to compare the cell means:

The critical t value at $p = .05$ for this df would be found in a t-table, then applied to the formula above using the **MSError** and the two relevant cell sizes. In a balanced design where both cells have the same size, the formula simplifies to

$$t\text{-critical} \times \sqrt{2 \times MS_{Error} / n}.$$

Reporting the Results

Sample APA Write-Up

A 2 x 2 factorial ANOVA was conducted to examine how box office category (\$500M+ vs. Under \$500M) and budget category (Over \$230M vs. Under \$230M) affect audience ratings of Superman movies. The cell means were: Under \$500M BO/Under \$230M Budget ($M = 86.00, n = 4$), Under \$500M BO/Over \$230M Budget ($M = 60.00, n = 1$), \$500M+ BO/Under \$230M Budget ($M = 82.50, n = 2$), \$500M+ BO/Over \$230M Budget ($M = 75.00, n = 1$). The main effect of box office category was not significant, $F(1,4) = 1.71, p = .261$. The main effect of budget category was significant, $F(1,4) = 14.51, p = .019$. The interaction between box office and budget category was not significant, $F(1,4) = 4.43, p = .103, MSe = 28.12$.

Key Reporting Elements

When reporting 2x2 factorial ANOVA results, be sure to communicate:

- Descriptive statistics for each cell (M, SD, n)
- Both main effects (F, df, p)
- The interaction effect (F, df, p, MSe)
- The direction of significant effects
- If the interaction is significant, describe the pattern of cell means
- Whether results support the research hypothesis